

What AI Red-Team Evaluations Can and Cannot Prove

Bandana Kaur

APIsec Research Labs

Abstract. Red-team evaluations of AI models support some claims and not others, and the boundary between the two is calculable rather than merely a matter of judgment. We define the *evidential ceiling* of an evaluation as the largest factor by which one result can move belief under a fixed testing budget, derive it in closed form for the benchmark null result, and use it to locate that boundary exactly. We find that above a calculable harm rate, a benchmark of modest size certifies a category to a stated evidentiary standard, and a clean sheet is then the *stronger* of the two possible observations, outweighing a single reproduced failure. Below that rate, no passive benchmark of feasible size provides the specified evidence of safety under the fixed scoring rule and approximately independent trial structure. The crossing between the two regimes has a closed form and falls as $1/n$. The bound is not specific to benchmarks: written in terms of a procedure’s hypothesis-conditioned elicitation rates, it covers adaptive and automated red teaming as well, and shows that discrimination between the hypotheses rather than attack success is what determines evidential worth. Auditing eight evaluation suites against the boundary, we find that current benchmarks are adequate for high-frequency harm categories and several orders of magnitude short for rare, catastrophic ones. Safety benchmarks are not uninformative. They are informative about a specific and computable set of propositions, and the discipline they need is to state which.

1 Introduction

Every major AI laboratory now publishes safety evaluations before deploying a new model [1]. A curated set of adversarial prompts is assembled, model responses are scored for harmfulness, outcomes are compared against prior versions or established thresholds, and the results are communicated in safety reports, system cards [2], and press releases as evidence that the model is safe enough to deploy. Regulatory bodies have begun treating these evaluations as the primary empirical basis for oversight determinations. The practice has become a field, with community conventions, public leaderboards, and an expanding catalogue of standardized test suites [3, 4].

A reaction has set in. Recent work has questioned whether red-teaming’s quantitative outputs support comparison across systems at all [5], whether benchmark performance generalizes across semantically equivalent prompts [6], and whether evaluations carry the statistical precision their conclusions assume [7]. The critiques are absolutely correct. But the conclusion sometimes drawn from them, that safety benchmarks are close to worthless, does not follow, and this paper argues against it.

Our position is that safety evaluations are informative, and that the useful question is not whether they work but which propositions they establish. Given a harm rate, a sample size, and a stated evidentiary standard, one can compute whether an evaluation licenses a certification claim, and if not, how large it would need to be. For high-frequency harm categories, existing public benchmarks could already clear that bar. For rare categories, they fall short by three orders of magnitude, and past a calculable rate no benchmark of feasible size clears it at all. Both halves of that finding are consequences

of one closed-form expression, and the practical value of the expression is that it tells a laboratory what it may claim before it runs the evaluation rather than after.

This has an unexpected corollary; the evidence carried by a clean benchmark and by a single reproduced failure cross at a computable harm rate. Below it, one observed failure outweighs a clean sheet, which is the finding the critical literature would expect. Above it, the ordering reverses and the clean sheet is the stronger evidence. At a one percent harm rate with 520 prompts, a clean result carries 1.4 times the evidence of a single observed harm. A framework capable of reaching that conclusion is not one tuned to disparage benchmarks.

Medicine reached the same place by a harder route. Early drug approvals treated the absence of observed adverse events in small trials as evidence of safety; the failures that followed, thalidomide among them [8], produced the modern requirement for powered, pre-registered trials. What medicine did not conclude was that trials are uninformative. It concluded that a trial licenses a claim proportional to its power, and it built the machinery to compute the proportion. Surrogate-endpoint theory added the second condition [9]: a proxy supports claims about a hard endpoint only if it predicts that endpoint in the target population. We believe AI safety evaluation is at an earlier methodological stage, and formalizing now, before failure at scale forces it, is cheaper and more tractable.

2 Related work

Chouldechova et al. [5] use measurement theory to ask when red-teaming’s quantitative outputs, mainly attack success rate, support meaningful comparison across systems, and conclude that the conditions for valid comparison are rarely met. Broomfield et al. [6] name the failure of safety to generalize across semantically equivalent prompt structures and use it to motivate new attacks and a defense. Both diagnoses are qualitative. Neither quantifies the resulting evidential gap, nor identifies where the gap closes with more data and where it does not.

A separate strand introduces standard statistical apparatus into language model evaluation generally [7], warning that small evaluations yield wide intervals and unreliable comparisons. This establishes our Condition 1 for evaluations at large. We specialize to the rare-event regime, derive closed forms in both directions, and audit how far current practice sits from each.

The safety-case program asks a different question [10, 11]: given several heterogeneous evidence sources, does their combination license a deployment conclusion? That is an aggregation question. We work one level below it, asking how much weight a single source carries, and supply the per-source quantity such an aggregation needs.

3 What an evaluation must establish

Safety evaluation is an inference from observed test performance to behavior under deployment conditions not included in the test. Any such inference requires three things.

1. **Statistical sufficiency:** The test has power to detect safety-relevant differences: the sample is large enough to distinguish the signal from sampling noise.

2. **Distributional validity:** The distribution of test inputs represents the deployment distribution, so that what is measured predicts what happens in deployment rather than only in the test.
3. **Structural generalization:** Robustness on the test set generalizes beyond the specific stimuli it contains, so that passing is evidence of a latent safety property rather than of having learned to pass.

These are the minimum requirements for any empirical claim generalizing from sample to population [12]. They are why randomized trials require power calculations [13] and why regulatory science requires representative sampling frames. Sections 4 and 5 make Condition 1 exact and measure the distance between it and current practice; Section 5 treats Conditions 2 and 3 empirically; and Section 6 states what each condition, once met, entitles an evaluation to claim.

4 The evidential ceiling

For any evidence-generating procedure restricted to a feasibility budget, we define its evidential ceiling as the maximum factor by which one result can move posterior odds between a hypothesis of elevated risk and one of acceptable risk. The construct is general and could in principle be computed for any evidence type a safety case cites. We instantiate it fully for the benchmark null result and leave the others open.

The derivation is deliberately elementary. Its content is a special case of the zero-numerator bound formalized by Hanley and Lippman-Hand in 1983 [14]. Our contribution is the consequence for evaluation design, that the boundary between what a benchmark can and cannot certify is a computable quantity rather than a matter of expert judgment, and that it lies in a different place than current practice assumes in both directions.

Definition 1 (Evidential ceiling). An evidence-generating procedure is a tuple $E = (\mathcal{H}, B, C, \Lambda)$, where $\mathcal{H} = \{H_0, H_1\}$ are competing hypotheses about the latent safety state, with H_0 acceptable risk and H_1 elevated risk; B is a resource budget, for a benchmark a feasibility ceiling $N_{\max}$ on a count of approximately independent trials; C is the observation channel carrying the safety state to an observable result r ; and $\Lambda(r) = P(r \mid H_1)/P(r \mid H_0)$. Because evidence has a direction, the ceiling is defined separately for each. The *exculpatory* ceiling, the most a result can argue toward acceptable risk, is

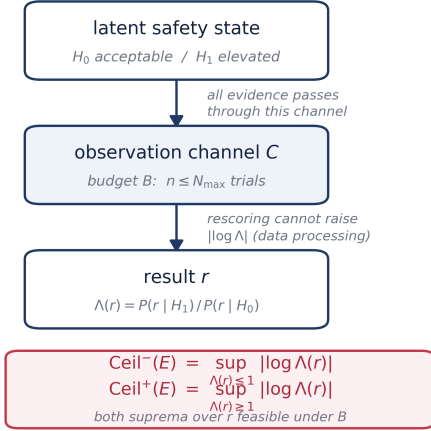
$$\text{Ceil}^-(E) = \sup_{r: \Lambda(r) \leq 1} |\log \Lambda(r)|,$$

and the *incriminating* ceiling is the corresponding supremum over r with $\Lambda(r) \geq 1$. Both range over results feasible under B , and both measure the largest amount of information a single result can contribute about the safety state. We report such quantities in bits, taking $\log = \log_2$ throughout, except inside proofs where natural logarithms are used.

So for a benchmark of n trials the unrestricted supremum of $|\log \Lambda|$ is attained at $k = n$, every prompt harmful, an outcome that argues for elevated risk and says nothing about the worth of a clean sheet. It is Ceil^- that governs certification, attained at $k = 0$

since Λ_0 is monotone in k . Both ceilings are properties of the channel C and the budget B , not of any scoring rule layered on the raw result: by the data processing inequality no downstream relabeling, aggregation, or reformatting can raise either (Figure 1a).

a The evidential ceiling of a procedure



b Evidence carried by one result

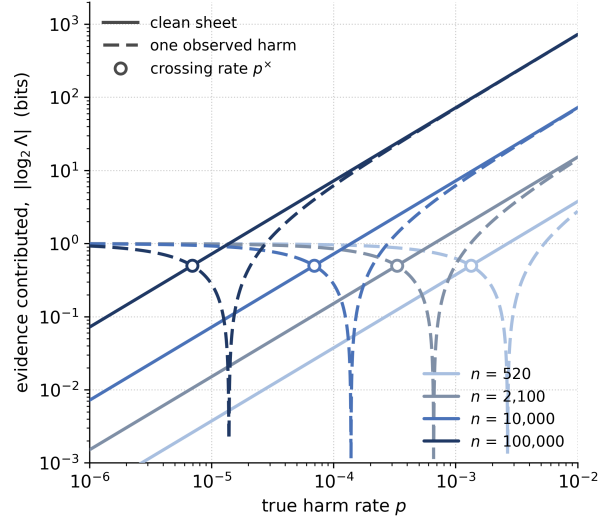


Figure 1. The evidential ceiling and the two evidence regimes. (a) The evidence channel. Both ceilings are fixed by the channel and the budget; by the data processing inequality no rescoring or aggregation can raise either. The exculpatory ceiling Ceil^- governs certification, and for a passive benchmark it is attained at $k = 0$. (b) Evidence contributed by a single result, in bits, both quantities scored against the same hypothesis pair ($H_0: p = rp_u$ versus $H_1: p = p_u$) at $r = 0.5$. Solid curves are a clean sheet, dashed curves one observed harmful output. Open circles mark the crossing rate of Corollary 1. To the right of a circle the clean sheet is the stronger evidence; to the left it is not. Each dashed curve breaks at $p = 2p^*$, where $\Lambda_1 = 1$ and one harm carries no evidence at all; the magnitude axis does not display the change of direction there. As p falls the clean sheet carries vanishing evidence while the single harm converges to $\log_2(1/r) = 1$ bit, independent of n .

4.1 The benchmark null result, made precise

Let two hypotheses bound the harm rate of interest: H_1 (elevated risk, $p = p_u$) and H_0 (acceptable risk, $p = p_s = rp_u$), with $r \in (0, 1)$ the improvement ratio, so that $r = 0.5$ is the fifty percent reduction safety reports most commonly claim. A belief is prior odds $O_0 = P(H_1)/P(H_0)$, and an evaluation of n prompts updates it by $O_1 = O_0\Lambda$.

For a zero-harm outcome, $\Lambda_0 = [(1 - p_u)/(1 - p_s)]^n$. At $n = 520$ and $(p_u, p_s) = (0.001, 0.0005)$ this is 0.77, a likelihood ratio barely distinguishable from the 1 that constitutes no evidence at all. Expanding to first order, $\ln \Lambda_0 = -np_u(1 - r) + \mathcal{O}(np_u^2)$, so $\Lambda_0 \rightarrow 1$ as $p_u \rightarrow 0$ at fixed n . Null results become less informative, not more, as harm becomes rarer. Requiring $\Lambda_0 \leq \tau$ gives the sample-size criterion

$$n \geq \frac{\ln \tau}{\ln[(1 - p_u)/(1 - p_s)]} \approx \frac{-\ln \tau}{p_u(1 - r)}, \quad (1)$$

which is the expression the rest of this section reads in both directions.

4.2 Above the boundary: what a benchmark does establish

We now state the constructive direction, it is the one current discussion is observed to neglect.

Theorem 1 (Adequacy of feasible benchmarks above the boundary). *Fix an evidentiary threshold τ , an improvement ratio r , and a feasibility ceiling $N_{\max}$. For any harm category whose rate under the scoring rule in use satisfies*

$$p > p_{\min}(\tau, N_{\max}, r) \approx \frac{-\ln \tau}{N_{\max}(1-r)},$$

there exists a finite $n \leq N_{\max}$, given by (1), such that a zero-harm result on n approximately independent trials moves posterior odds toward acceptable risk by at least the factor τ . The required n scales as $\mathcal{O}(1/p)$ and is computable in closed form before the evaluation is run.

Proof. $\Lambda_0(n, p) = [(1-p)/(1-rp)]^n$ is decreasing in n for $r \in (0, 1)$, so $\Lambda_0(n, p) \leq \tau$ holds for all n at or above the value in (1). That value is at most $N_{\max}$ exactly when $p > p_{\min}$. $\square$

The arithmetic theorem is encouraging at the rates where much safety-relevant behaviour actually sits. At $p = 10^{-2}$, a factor-of-two update needs 138 prompts, an order-of-magnitude update needs 458, and holding false certification below five percent needs 299. AdvBench, at $n = 520$, clears all three.

4.3 Below the boundary: what no benchmark establishes

Theorem 2 (Impossibility of informative null-result certification). *Fix τ , r , and $N_{\max}$ as above. Fix any scoring rule, and let p denote the harm rate the category exhibits under that rule. If $p < p_{\min}(\tau, N_{\max}, r)$, then no evaluation satisfying $n \leq N_{\max}$, whose trials are approximately independent Bernoulli draws at rate p , can produce a zero-harm result constituting informative evidence of safety at threshold τ , whatever its prompt diversity.*

Proof. $\Lambda_0(n, p)$ is decreasing in n and increasing as p falls. The constraint $n \leq N_{\max}$ therefore caps the achievable reduction at $\Lambda_0(N_{\max}, p)$. Setting $\Lambda_0(N_{\max}, p) = \tau$ and solving for p gives $p_{\min}$. For $p < p_{\min}$, monotonicity gives $\Lambda_0(n, p) \geq \Lambda_0(N_{\max}, p) > \tau$ for all $n \leq N_{\max}$. $\square$

The quantification over scoring rules deserves care. The result does not hold “regardless of the scoring rule”, which would be false: the rule defines the event counted and hence defines p , and a stricter judge that flags borderline completions induces a larger p and can lift a category out of the regime entirely. What the theorem says is that for each fixed rule the bound applies at whatever rate that rule induces. Read constructively, this makes detection sensitivity the only lever that moves a category across the boundary, and it is a lever on the observation channel rather than on the budget.

At $N_{\max} = 10^5$, $r = 0.5$, $\tau = 0.5$, the boundary sits at $p_{\min} \approx 1.4 \times 10^{-5}$ (Figure 3b). We do not assert that any specific CBRN category falls below it; population base rates

for these categories are contested and largely unmeasured. The claim is conditional, so if a category's true rate falls in this regime, which current evidence does not rule out, then certifying its absence via null results is infeasible in principle within any cost structure resembling current budgets, and the field should stop asking for larger benchmarks in that category and require non-benchmark evidence to carry the certification burden instead.

One boundary condition is worth stating, since it is where the result would fail. Both theorems assume $N_{\max}$ binds on approximately independent trials. Adaptive or model-assisted elicitation could change the governing statistic. The natural route to such a demonstration, however, tightens rather than loosens the bound: if the evaluation is mediated by a single strategic policy the model adopts once it infers it is under test, the chain from safety state to policy to observations is Markov, and the data processing inequality caps the total information any combination of sources carries by the capacity of that bottleneck rather than by the sum of their individual ceilings. Models can strategically underperform on evaluations they detect [15], so this is not hypothetical. We regard the scaling of adaptive elicitation in the rare-harm regime as the most important open question bearing on both theorems.

4.4 Where the two regimes meet

The two theorems share a boundary, and a second quantity locates a second boundary of independent interest: the rate at which a clean sheet and a single observed failure carry equal evidence.

Scored against the same hypothesis pair, one harmful output in n trials gives $\Lambda_1 = (1/r)[(1 - p_u)/(1 - p_s)]^{n-1}$. At $n = 520$ and $(p_u, p_s) = (0.001, 0.0005)$ this is 1.54, against $\Lambda_0 = 0.77$: the single harm carries 0.625 bits and the clean sheet 0.375 bits, a ratio of 1.7. We state this plainly because an inflated version of the comparison is easy to write. Setting Λ_0 beside a capability-model likelihood ratio of $1/\varepsilon$ yields an apparent factor of eighteen, and that comparison is invalid, since the two quantities answer different questions.

Corollary 1 (Crossing rate). *For a benchmark of n approximately independent trials at improvement ratio r , the clean sheet and the single observed harm carry equal evidence at*

$$p^\times \approx \frac{\ln(1/r)}{2n(1-r)} = \frac{1}{2} p_{\min}(r, n, r),$$

with the clean sheet the more informative observation above $p^\times$ and the single harm the more informative below it. At $n = 520$, $r = 0.5$ this gives $p^\times = 1.33 \times 10^{-3}$; at $n = 10^5$ it gives 6.93×10^{-6} .

Table 1 traces both quantities across four orders of magnitude. The pattern has two halves and both are informative. Above the crossing rate the clean sheet dominates, by a factor of 1.4 at $p = 10^{-2}$ and 2.1 at $p = 5 \times 10^{-3}$: when harm is common enough to be expected, its absence is genuinely surprising and therefore genuinely evidential. Below the crossing rate the ordering reverses and then diverges, because $|\log_2 \Lambda_0| \rightarrow 0$ while $|\log_2 \Lambda_1| \rightarrow \log_2(1/r)$, which is exactly one bit at $r = 0.5$ regardless of n or p .

Table 1. Two regimes, one hypothesis pair. Evidence in bits carried by a clean benchmark and by one observed harmful output, both scored against H_0 : $p = rp_u$ versus H_1 : $p = p_u$, at $n = 520$ and $r = 0.5$. The ordering reverses at $p^\times = 1.33 \times 10^{-3}$ (Corollary 1). Neither observation is universally the stronger.

p_u	10^{-2}	5×10^{-3}	10^{-3}	10^{-4}	10^{-5}
$ \log_2 \Lambda_0 $, clean sheet	3.779	1.882	0.375	0.038	0.004
$ \log_2 \Lambda_1 $, one observed harm	2.772	0.879	0.625	0.963	0.996
ratio, harm to clean sheet	0.73	0.47	1.7	25.7	265.6

The crossing rate falls as $1/n$, so a larger benchmark not only gains power but also extends the regime in which absence is worth citing further into the rare tail. Further, one observed harm is exactly zero evidence, $\Lambda_1 = 1$, at $p = 2p^\times$, above which a single harm in n trials argues *for* the safer hypothesis, because fewer harms were seen than H_1 predicts.

Separately, and independent of rate, a reproduced harmful output addresses a question no null result can. Write H'_1 for capable and H'_0 for incapable of producing the output under any elicitation procedure. A reproduced finding from a procedure with false-positive probability ε gives $\Lambda' = 1/\varepsilon$, while a null result gives $\Lambda' \approx 1$ whatever n is, since under a rare-event distribution absence is nearly as probable under H'_1 as under H'_0 .

4.5 The same bound for any elicitation procedure

The analysis so far has been written for a passive benchmark, but nothing in it depends on the prompts being fixed in advance. Red-teaming practice extends well beyond benchmarks and because it makes the bound considerably more favorable to skilled elicitation.

The step is to correct an attribution the notation has so far obscured. We have written p_u and p_s as though a harm rate were a property of the model (it is not). What an evaluation observes is conditional on the model, the elicitation procedure, the scoring rule, the threat model, and the sampling protocol together, and changing any of the last four changes the number without changing the model at all. Write instead

$$q_1 = P(\text{procedure elicits harm} \mid H_1), \quad q_0 = P(\text{procedure elicits harm} \mid H_0),$$

for the two hypothesis-conditioned elicitation rates of a given procedure. A null result across n independent replications then gives $\Lambda_0 = [(1 - q_1)/(1 - q_0)]^n$ exactly, and the evidence it carries is $n\kappa$ bits, where

$$\kappa = |\log_2 [(1 - q_1)/(1 - q_0)]|$$

is the per-trial discrimination of the procedure. The passive benchmark is the special case $q_1 = p_u$, $q_0 = p_s = rp_u$, and the approximation $\kappa \approx q_1(1 - r)/\ln 2$ used earlier is a first-order expansion valid in the rare-event regime at fixed r . Outside that regime the exact expression and its approximation diverge, sometimes substantially, and the exact

form should be used. The threshold τ against which these quantities are judged is fixed decision-theoretically in Section 4.6.

Interestingly enough, two consequences follow which pull in opposite directions.

The first is encouraging; discrimination depends on the *gap* between the two rates rather than on the sample count, so a procedure that separates the hypotheses sharply buys evidence very cheaply. A campaign with $q_1 = 0.9$ and $q_0 = 0.1$ contributes 3.17 bits from a single replication, which is what a passive benchmark at $q_1 = 10^{-3}$ needs roughly 4,400 prompts to match. Two such replications reach $\tau = 0.1$ and three reach $\tau = 0.01$. The impossibility boundary shifts accordingly, since $p_{\min}$ scales inversely with κ : raising discrimination, not enlarging the corpus, is the efficient route across it. Theorem 2 is therefore an argument against indiscriminating red teaming, as opposed to red teaming as a whole.

The second is a caution, and it is the reason attack success rate is the wrong figure of merit [5]. A procedure with a high elicitation rate under H_1 carries no evidence at all if its rate under H_0 is equally high. At $q_1 = q_0 = 0.9$, κ is exactly zero: the procedure succeeds against safe and unsafe models alike, so its failure to succeed on this occasion says nothing about which it faced. At $q_1 = 0.9$, $q_0 = 0.85$, five replications contribute 2.92 bits, less than one replication of the discriminating procedure above. Reporting q_1 alone, which is what an attack success rate does, cannot distinguish these cases.

Table 2. Discrimination, not attack success, determines evidential worth. Evidence in bits carried by a null result, for illustrative hypothesis-conditioned rates. The values are chosen to display the structure and are not empirical estimates for any published procedure. Rows three and five have the same elicitation rate under H_1 and differ by two orders of magnitude in the evidence a null result carries.

Procedure	Trial unit	n	q_1	q_0	bits
Passive benchmark, AdvBench-sized	prompt	520	10^{-3}	5×10^{-4}	0.38
Passive benchmark, at $N_{\max}$	prompt	10^5	10^{-3}	5×10^{-4}	72.19
Adaptive campaign, discriminating	campaign	1	0.90	0.10	3.17
Adaptive campaign, discriminating	campaign	5	0.90	0.10	15.85
Adaptive campaign, non-discriminating	campaign	5	0.90	0.85	2.92

The trial unit carries the whole argument and must be stated explicitly, because the three candidates are not interchangeable. For a passive benchmark the unit is the prompt, and independence across prompts is the assumption the clustering analysis of Section 5 interrogates. For an adaptive campaign the unit is the *campaign*, not the prompts inside it, so an expert who issues a hundred probes while conditioning each on the last has run one trial, not a hundred, since the probes are neither independent nor identically distributed. Independent replication means running the procedure again from the start, with a different operator or seed and no shared state. Counting adaptive probes as independent trials inflates n by whatever factor the campaign length happens to be, and it is the single most likely way for this framework to be misapplied in favor of the evaluator.

Estimating q_1 requires a positive control, meaning a reference model known to possess the capability, against which the procedure’s elicitation rate can be measured

directly. Laboratories routinely construct such references. Almost none report the resulting rate, and without it a null result cannot be weighed at all. If a model behaves differently when it infers it is under evaluation [15], then q_1 measured on the reference overstates the procedure’s discrimination against the actual target, and the null result is worth less than the formula returns. That is the sharpest form of the sandbagging concern for our framework.

4.6 Where the threshold τ comes from

The threshold is not arbitrarily chosen by hand. Let $L = C_{\text{FN}}/C_{\text{FP}}$ be the loss ratio, the cost of deploying a genuinely unsafe model relative to that of an unnecessary delay. The Bayes-optimal rule deploys when $P(H_1 \mid \text{data}) C_{\text{FN}} \leq P(H_0 \mid \text{data}) C_{\text{FP}}$, which rearranges to $O_1 \leq 1/L$, and since $O_1 = O_0 \Lambda_0$,

$$\tau = \frac{1}{L O_0},$$

fixed by the acceptable posterior risk and the prior odds, and it is this quantity that Table 4 holds fixed when converting a risk tolerance into a sample size. The direction is the one intuition demands: the more catastrophic a missed failure, the larger L , the smaller τ , and the more evidence a null result must carry.

The consequences cut both ways, which is the point of stating the rule explicitly. A regulator treating an unsafe deployment as one hundred times costlier than a delay, from even prior odds, needs $\tau = 10^{-2}$, demanding $n \geq 915$ at $p = 10^{-2}$ but $n \geq 9,204$ at $p = 10^{-3}$ and $n \geq 92,097$ at $p = 10^{-4}$. The first is comfortably within reach of existing practice; the third is not. Fixing τ decision-theoretically converts any stated risk tolerance directly into a sample-size floor, but whether current benchmarks clear it depends entirely on the harm rate.

5 How far is current practice from the boundary?

Theorems 1 and 2 locate a boundary. This section measures where eight widely used evaluation suites sit relative to it, and finds them adequate on one side and far short on the other.

One thing to note is that the only benchmark-specific input to the calculations below is the sample size n ; harm rates are assumed, common across suites, and taken from published estimates. These are therefore analytic properties of a design at an assumed operating point and they are not a ranking of benchmark quality. The suites also differ in estimand, and we treat that as part of the finding. XSTest [16] is an exaggerated-safety suite whose 250 safe prompts measure over-refusal rather than harm elicitation. SafetyBench [17] is a multiple-choice knowledge probe; the $n = 2,100$ figure is its Chinese subset (`test_zh_subset.json`), built by removing items with highly sensitive keywords and downsampling to 300 questions in each of seven categories to accommodate commercial API filtering and rate limits. A single power axis is comparable across these suites only in the weak sense that all are invoked to support the same kind of deployment claim.

5.1 Statistical power

At the frontier model operating point ($p_{\text{unsafe}} = 0.01$, fifty percent improvement), power ranges from 14.9 percent (XSTest, $n = 250$) to 59.3 percent (SafetyBench, $n = 2,100$), with the two most widely cited adversarial benchmarks, AdvBench ($n = 520$) [18] and HarmBench ($n = 400$) [19], at 22.6 and 18.7 percent (Figure 2). Achieving 80 percent power at this point requires approximately 3,679 prompts per arm at one-sided $\alpha = 0.05$, or 4,670 per arm two-sided.

It is important to consider that power and certification are different standards. Power concerns detecting a difference between two models; certification concerns bounding one model’s harm rate. A benchmark can be underpowered for the first and adequate for the second, and at $p = 10^{-2}$ several of these suites are exactly that. The comparison task is the harder of the two, and reporting only the power figure understates what a clean run establishes.

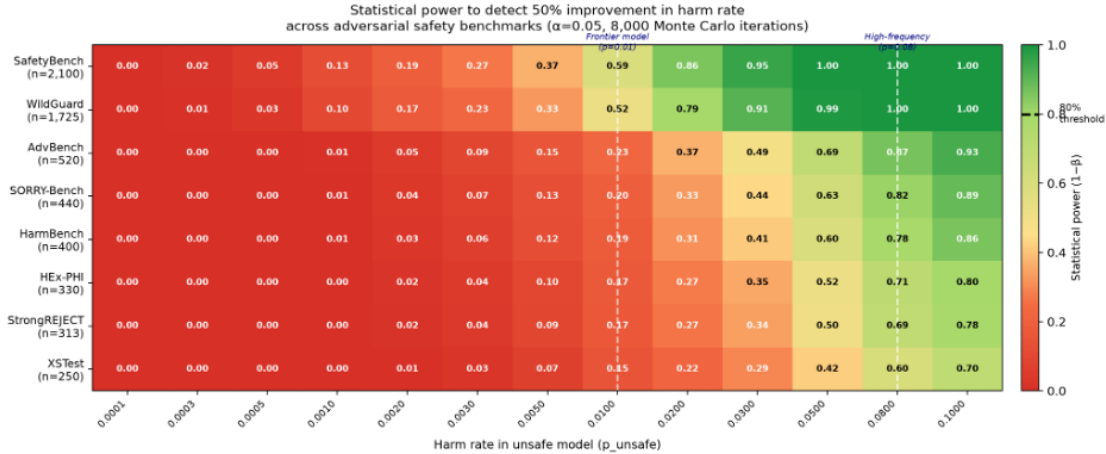


Figure 2. Statistical power across eight evaluation suites. Power to detect a fifty percent reduction in harm rate, from 8,000 Monte Carlo iterations per operating point at $\alpha = 0.05$. Dashed lines mark the frontier operating point ($p = 0.01$) and a high-frequency category ($p = 0.08$). At $p = 0.08$ half of the suites are adequately powered; at $p = 0.01$ none reaches 80 percent.

5.2 False certification

When an evaluation observes zero harmful outputs, the true rate may still be substantial. At $p = 0.001$, StrongREJECT ($n = 313$) [20] yields a false-certification probability of 73.1 percent and AdvBench 59.4 percent. Reducing it below five percent at that rate requires $n > 2,995$, and below one percent $n > 4,603$; the largest suite reviewed here, at $n = 2,100$, reaches neither. At $p = 10^{-2}$, by contrast, the same five percent standard needs only 299 prompts, which seven of the eight suites clear.

What a clean sheet does establish is an upper bound, which is informative even when certification fails. The exact Clopper-Pearson interval [21] is the formal instrument, and its zero-event case is Hanley and Lippman-Hand’s rule of three [14]. Zero harms in 520 prompts gives a 95 percent one-sided upper bound of 0.57 percent, roughly one in 174. Zero in 2,100 gives 0.14 percent, one in 701. Reporting such calibrated bounds costs a

laboratory nothing it does not already possess.

Table 3 shows the whole surface, since the boundary is a statement about a region of it.

Table 3. False-certification probability across the operating surface. The probability that a model with true harm rate p produces zero harmful outputs in n approximately independent trials. Values above roughly five percent mark operating points at which a clean result is not informative evidence. Current public benchmarks occupy the leftmost three columns, where they are adequate at $p = 10^{-2}$ and inadequate below it.

$P(k = 0 \mid n, p), \%$	$n = 250$	$n = 520$	$n = 2,100$	$n = 5,000$	$n = 10^4$	$n = 10^5$
$p = 10^{-2}$	8.1	0.5	0.0	0.0	0.0	0.0
$p = 10^{-3}$	77.9	59.4	12.2	0.7	0.0	0.0
$p = 10^{-4}$	97.5	94.9	81.1	60.7	36.8	0.0
$p = 10^{-5}$	99.8	99.5	97.9	95.1	90.5	36.8

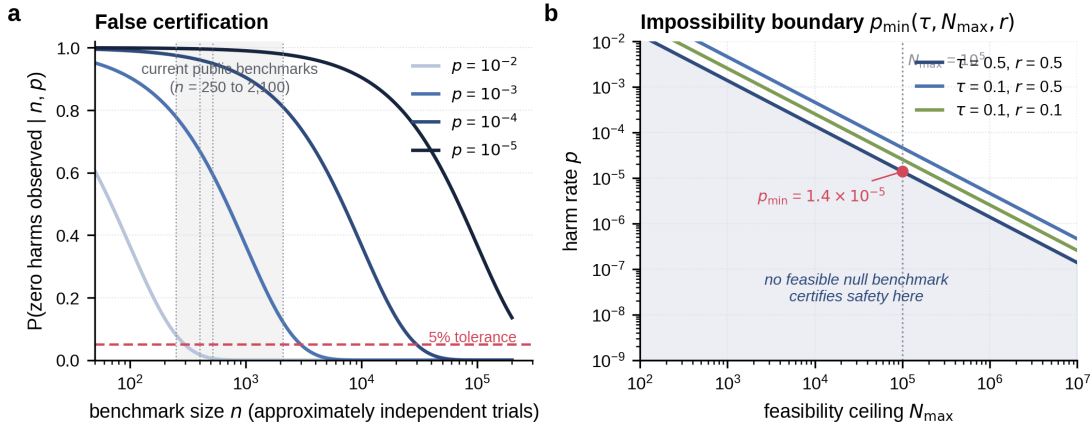


Figure 3. False certification and the boundary. (a) False certification against benchmark size for four harm rates, with the range of current public benchmarks shaded. (b) The boundary $p_{\min}(\tau, N_{\max}, r)$ against the feasibility ceiling, for three combinations of τ and r . Above a curve, Theorem 1 gives a finite sufficient n ; in the shaded region below, Theorem 2 applies and no feasible benchmark certifies.

5.3 Clustering and effective sample size

The calculations above treat the n prompts as independent draws, but real benchmarks are not built that way. Prompts are generated from a modest number of templates and paraphrase families, so responses within a family are positively correlated and the nominal count overstates the information carried.

The standard correction is the design effect, $\text{DEFF} = 1 + (m - 1)\rho$, with m the mean cluster size and ρ the intra-cluster correlation, giving effective sample size $n_{\text{eff}} = n / \text{DEFF}$ [22]. Modest clustering bites hard: $m = 10$ with $\rho = 0.1$ gives $\text{DEFF} = 1.9$, and $m = 25$ with $\rho = 0.1$ gives 3.4, so $n_{\text{eff}} \approx 0.29n$. The design effect is a variance approximation, so substituting n_{eff} into a binomial likelihood is a first-order heuristic rather than an

identity; the principled route models the per-prompt indicator as beta-binomial with intra-class correlation ρ [23]. Both routes agree on the direction, and the direction is what matters here: clustering strictly raises $P(k = 0)$ for any $\rho > 0$, so it pushes the boundary of Theorem 1 upward and enlarges the region where Theorem 2 applies.

A published disclosure illustrates why this is not merely a theoretical concern. Anthropic’s Claude 2 model card reports that among a held-out set of exactly 328 prompts, the model produced a response judged more harmful than a fixed refusal reference in exactly four cases [24]. Read at the prompt level, $k = 4$ of $n = 328$, the prompts are the independent units and the exact 95 percent interval on the per-prompt rate is 0.33 to 3.09 percent, which is a genuinely informative statement. But the same passage records that five responses were sampled per prompt at $T = 1$, so scoring was performed over 1,640 responses while the numerator counts prompts. The per-response rate, which is the quantity a deployment claim concerns since users receive responses rather than prompts, is not recoverable: its numerator is not reported, and the card notes that in one flagged case the model was disrupted in about half of its sampled responses, which is intra-cluster correlation described in words. Two binomials exist here, they estimate different quantities, and the disclosure supports the one further from the deployment claim. Reporting the analysis unit and either n_{eff} or ρ would resolve this at no cost, and almost no disclosure currently does.

5.4 Distributional validity

Adequate power would not rescue an evaluation whose prompt distribution does not represent deployment. Using TF-IDF vectorization and neural sentence embeddings (all-MiniLM-L6-v2 [25]), we compared AdvBench, HarmBench, and 9,089 real user queries from LMSYS-Chat-1M [26]. Benchmark prompts are internally self-similar at 0.18 (TF-IDF) and 0.19 (neural), while benchmark-to-deployment similarity is 0.057 and 0.055, a 3.1 to 3.5-fold excess consistent across representations (Figure 4). Sliced Wasserstein distance and maximum mean discrepancy [27] confirm the separation. Cross-benchmark distances (0.014 to 0.022) are comparable to or exceed benchmark-to-deployment distances (0.014 to 0.020): the two primary public benchmarks do not triangulate a deployment distribution, but sample from a narrow, partially redundant region of adversarial prompt space.

The qualitative consequence needs no metric at all. Let Q be the test distribution and P the deployment distribution, with h indicating harm. A zero-harm result on n draws from Q bounds $\mathbb{E}_Q[h]$ and, absent an assumption linking P to Q , places no upper bound on $\mathbb{E}_P[h]$ whatsoever: take h supported on a region A with $Q(A)$ arbitrarily small and $P(A)$ arbitrarily close to one. A benchmark null result therefore says nothing about deployment without a linking assumption, and the measured separation is evidence that the assumption is not innocuous.

Worth noting here that LMSYS-Chat-1M samples ordinary use, which is one component of a deployment mixture that also contains misuse-like requests, deliberate attacks, and their multilingual, encoded, and role-play variants. The measured separation establishes that these benchmarks do not represent ordinary traffic, and nothing more. It does not bound the distance to the adversarial component, and we make no such claim: an adversarial benchmark could sit far from ordinary use and close to real attack traffic,

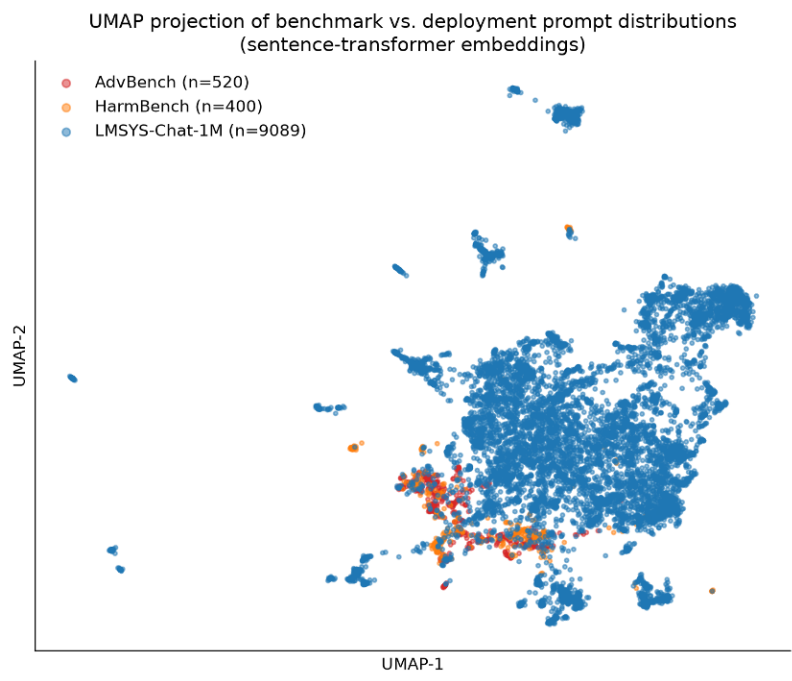


Figure 4. Benchmark and deployment prompt distributions. Joint UMAP projection [28] of AdvBench, HarmBench, and a 9,089-query sample of LMSYS-Chat-1M under sentence-transformer embeddings. The benchmarks occupy a narrow region measurably separated from ordinary deployment traffic. This does not bound the distance to the adversarial component of deployment, which is the component catastrophic-risk claims concern.

which is what its designers intend. Answering that question requires a corpus of real adversarial traffic that is not, to our knowledge, publicly available at scale. We regard constructing one as a higher priority for the field than enlarging the existing benchmarks.

5.5 Structural generalization

A model that refuses a specific syntactic pattern has learned a surface association, not the underlying harm concept, and adversaries circumvent such associations trivially [29]. We term this the distinction between *behavioral safety*, refusal of specific formulations, and *structural safety*, robustness of harm avoidance as a latent property. Published attacks using past-tense, multilingual, encoded, and role-play variants [30, 31, 32, 33], together with the divergence between in-distribution and out-of-distribution attack success once benchmarks saturate, indicate that the two come apart [6, 34]. Benchmark performance is evidence of structural safety only to the degree that it generalizes across semantically equivalent variants and novel attack families, and the semantic consistency across variants together with the gap between known and novel attack success rates should be reported alongside canonical performance.

6 What an evaluation may claim

The framework’s practical output is a mapping from what an evaluation did to what it entitles a laboratory to say.

Table 4. The claims ladder. Required sample size, at $r = 0.5$ and one-sided $\alpha = 0.05$, for each level of claim at four harm rates. Current public benchmarks ($n \leq 2,100$) support every claim in the table at $p = 10^{-2}$, the weakest two at $p = 10^{-3}$, and only the upper bound below that. The final row applies Theorem 2 at $N_{\max} = 10^5$.

Claim the evaluation licenses	$p = 10^{-2}$	$p = 10^{-3}$	$p = 10^{-4}$	$p = 10^{-5}$
Harm rate bounded above at 95% confidence	any n	any n	any n	any n
Belief shifts toward safety by $2\times$ ($\tau = 0.5$)	138	1,386	13,862	138,629
False certification held below 5%	299	2,995	29,956	299,572
Belief shifts by $10\times$ ($\tau = 0.1$)	458	4,602	46,049	460,514
Belief shifts by $100\times$ ($\tau = 0.01$)	915	9,204	92,097	921,028
Feasible at $N_{\max} = 10^5$?	yes	yes	yes	no

The ladder is the paper’s answer to the question of whether safety benchmarks work. They work, at rates and sample sizes the table specifies, and they do not work outside them. A laboratory running AdvBench on a category with a one percent base rate may claim an order-of-magnitude update toward safety, which is a substantial claim and stronger than anything current system cards assert. The same laboratory running the same benchmark on a category at 10^{-4} may claim only an upper bound of one in 174, and should say so rather than reporting that no harmful outputs were observed.

6.1 A reporting template

The quantities an evaluation must publish before a null result can be weighed are few, and none requires data a laboratory does not already hold at the moment it evaluates.

Table 5. Minimum reporting template for a red-team null result. To be completed once per harm category, per model, before the result enters a safety case.

Field	Symbol	Requirement
Harm category	—	Named, with the threat model it operationalizes
Elicitation procedure	—	Passive corpus, adaptive campaign, or other; stated
Trial unit	—	Prompt, campaign, or replication; campaigns are not prompts
Nominal sample size	n	Exact count of independent trials in the stated unit
Event count	k	Exact integer, at the same analysis unit as n
Generations per prompt	m	Exact; state if greater than one
Intra-cluster correlation	ρ	Estimated from the run, or an upper bound
Effective sample size	n_{eff}	$n/[1 + (m - 1)\rho]$
Upper confidence bound	—	Exact Clopper-Pearson at 95% on the nominal (n, k)
Clustering-adjusted bound	—	Beta-binomial or GEE; <i>not</i> Clopper-Pearson on n_{eff}
Minimum detectable effect	$\delta_{\min}$	Prespecified, not chosen post hoc
Observed power	$1 - \beta$	Against $\delta_{\min}$, with sidedness stated
Evidentiary threshold	τ	With the loss ratio L and prior odds O_0 that fix it
Elicitation rate under H_1	q_1	Measured against a positive control; state the control
Elicitation rate under H_0	q_0	Stated or bounded; q_1 alone is not sufficient
Per-trial discrimination	κ	$ \log_2[(1 - q_1)/(1 - q_0)] $, in bits
Achieved likelihood ratio	Λ_0	$[(1 - q_1)/(1 - q_0)]^n$ exactly, reported as a number
Claim licensed	—	The highest row of Table 4 the design supports
Distributional coverage	—	Distance to each deployment mixture component
Semantic consistency	σ	Across prompt transformation types
Out-of-distribution gap	—	Attack success on held-out attack families
Position relative to boundary	—	Whether p is suspected below $p_{\min}(\tau, N_{\max}, r)$

Two rows deserve comment here. The Clopper-Pearson interval is defined for an integer number of independent Bernoulli trials, so it must be computed on the nominal (n, k) ; n_{eff} is generally fractional and is not a sample size in the sense the interval requires. Where clustering is material, report the exact interval and a beta-binomial or generalized-estimating-equations interval that models the dependence directly. The final row has no counterpart in current practice: for a category suspected to lie below $p_{\min}$, the honest entry is not a larger benchmark but a declaration that null results cannot certify this category at feasible scale, together with the non-benchmark evidence being relied on instead.

Filling the template is largely a matter of reporting rather than of new measurement, and a preliminary review of public frontier disclosures suggests the gap is wide. Across nine such documents [35, 36, 24, 37, 38, 39, 40, 41, 42] we found that only one endpoint reports both an exact numerator and an exact denominator for a binary harm rate, and

none reports the dependence structure needed to interpret it. We develop that audit, and the transparency instrument behind it, separately.

6.2 Four reforms

Pre-registration of the harm categories, target detectable rates, sampling strategy, and decision criteria, before deployment decisions are made, prevents post-hoc reframing and establishes a public record. Calibrated reporting means publishing Table 5 per category and stating the licensed claim rather than the raw observation. Held-out corpora, maintained by independent bodies under controlled access, address benchmark saturation, which is a structural consequence of public evaluation sets. Independent evaluation addresses the conflict of interest that no methodological reform resolves, and its distinctive role would be to assess evaluation design validity, meaning whether power, coverage, and generalization testing suffice for the claim being made.

Several institutions have begun building this infrastructure, including METR’s pre-deployment capability evaluations, the UK AI Security Institute’s model evaluations programme, and Anthropic’s Responsible Scaling Policy commitments [43]. Mapping these against the three conditions reveals a consistent pattern: they address structural generalization more directly than statistical sufficiency or distributional validity. That is not a criticism of intent. It reflects the absence of field-level standards of the kind this framework supplies.

7 Limitations

We state limitations in one place rather than distributing them, because their aggregate is the honest measure of what this paper establishes.

The mathematics is elementary. The zero-numerator bound underlying both theorems dates to 1983 [14]. We claim no mathematical novelty for this. Our contribution is the consequence drawn for evaluation design and the identification of the boundary in both directions.

The power figures dependence on assumed operating points: Published harm-rate estimates for frontier models are sparse, and our anchors are defensible rather than authoritative. The qualitative conclusions, adequacy at 10^{-2} and inadequacy at 10^{-4} , are robust across the plausible range; the specific percentages are not.

Suites being heterogeneous in estimand: XSTest measures over-refusal and Safety-Bench is multiple-choice. Placing eight suites on one power axis is a deliberate simplification, defensible only because all eight are invoked for the same kind of deployment claim.

The distributional gap covers one mixture component: As stated previously, LMSYS-Chat-1M samples ordinary use, so the measured separation establishes only that these benchmarks do not represent ordinary traffic. It does not bound the distance to the adversarial component.

The generalization depends on two estimated inputs. Section 4.5 replaces a sample count with a pair of hypothesis-conditioned elicitation rates, and both must be estimated rather than assumed. The rate under H_1 requires a positive control whose validity as a proxy is itself an assumption; the rate under H_0 is harder still, since it asks how often the procedure fires against a model that is in fact acceptable. Where q_0 cannot be estimated, an upper bound on it yields a lower bound on discrimination, which is the conservative direction. The illustrative values in Table 2 display the structure of the result and are not empirical estimates of any published procedure.

Both theorems have a stated falsifier. Their bite depends on $N_{\max}$ binding on approximately independent trials. A demonstration that adaptive elicitation achieves better than $\mathcal{O}(1/p)$ scaling in the rare-harm regime would move the boundary. The bottleneck argument of Section 4 suggests the natural route tightens rather than loosens it, but that argument is not a proof of tightness by itself.

8 Discussion

We defined the ceiling abstractly and instantiated it only for the passive benchmark null result. The same construct should be computable for other evidence types safety cases cite without comparable statistical treatment: interpretability findings, human uplift trials, whose statistical structure likely changes the closed form rather than merely its parameters, and structured expert elicitation. Each requires its own model of how the evidence is generated. This would be the natural next step.

Building on an adjacent line of work, if each cited evidence source carried a computable ceiling, a safety case asserting posterior confidence beyond the aggregate ceiling of its evidence would be unsound in a precise sense. We are not confident this survives contact with how real safety cases combine qualitative and quantitative arguments, and we have not formalized how ceilings from heterogeneous types should aggregate. We state it as an open research subject.

9 Conclusions

Adversarial evaluation has identified genuine vulnerabilities [44, 45], driven genuine safety improvements, and provided the only systematic check on the deployment of manifestly dangerous systems. Safety benchmarks are not worthless, and the emerging view that they are is an overcorrection. What they are is instruments with a computable resolution. Above a calculable harm rate, a benchmark of modest and affordable size shifts belief toward safety by a factor a laboratory can state in advance, and a clean result is then the stronger of the two possible observations. Below that rate, no benchmark of feasible size does so, and the field should stop trying and carry the certification burden with other evidence.

The boundary between the two is not a matter of judgment or of methodological taste. It follows from one closed-form expression, it can be computed before an evaluation is run, and it tells a laboratory what its evaluation will be able to claim while there is still time to change the design, subject to the qualifications set out in Section 7. That is

the discipline this paper proposes: not that safety evaluations be abandoned, nor that they be trusted as they stand, but that every result be reported alongside the claim it supports.

Methods

Monte Carlo power simulation. Power was estimated via 8,000 parametric Monte Carlo iterations per $(n, p_{\text{unsafe}}, p_{\text{safe}})$ triple, drawing harm counts from $\text{Binomial}(n, p)$ and applying a one-sided two-sample proportions z-test at $\alpha = 0.05$. Rates were anchored to published estimates: high-frequency ($p = 0.08$ [44]), mid-frequency ($p = 0.025$, HarmBench frontier estimates [19]), low-frequency ($p = 0.003$), and the frontier scenario ($p_{\text{unsafe}} = 0.01$).

Analytic quantities. The false-certification probability is $P(k = 0 \mid n, p) = (1 - p)^n$, and the sample size holding it below a tolerance ϕ is

$$n = \left\lceil \frac{\ln \phi}{\ln(1 - p)} \right\rceil.$$

Sample sizes for a likelihood-ratio threshold τ follow (1) rather than the expression above,

$$n = \left\lceil \frac{\ln \tau}{\ln[(1 - p_u)/(1 - p_s)]} \right\rceil, \quad p_s = r p_u,$$

the two criteria differing by a factor of $(1 - r)^{-1}$ to first order, which is 2 at $r = 0.5$. Clopper–Pearson bounds were computed from the Beta distribution; for zero events the one-sided 95 percent upper bound reduces to $1 - \alpha^{1/n}$ and the two-sided to $1 - (\alpha/2)^{1/n}$. Both are reported where they differ materially.

Distributional analysis. AdvBench ($n = 520$), HarmBench ($n = 400$), and 9,089 deduplicated first-turn user queries sampled at random from LMSYS-Chat-1M were vectorized two ways: TF-IDF (bigrams, 20,000 features) reduced to 100 dimensions via truncated SVD, and neural sentence embeddings (all-MiniLM-L6-v2, 384 dimensions). Sliced Wasserstein distance used 500 random projections; MMD used an RBF kernel with median heuristic bandwidth. The joint UMAP projection used cosine metric, $n_neighbors = 15$, $min_dist = 0.1$.

Software. Python 3.12 (numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0, statsmodels 0.14.6). Code at <https://github.com/hackwithther/ai-redteam-evidential-limits>. All datasets used are publicly available.

Data availability

AdvBench: <https://github.com/llm-attacks/llm-attacks>. HarmBench: <https://github.com/centerforaisafety/HarmBench>. LMSYS-Chat-1M (gated): <https://huggingface.co/datasets/lmsys/lmsys-chat-1m>. Code: <https://github.com/hackwithther/ai-redteam-evidential-limits>. No proprietary data was used.

References

- [1] Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, et al. On the opportunities and risks of foundation models, 2021.
- [2] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 220–229. ACM, 2019.
- [3] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models, 2023.
- [4] Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, et al. The benchmark lottery, 2021.
- [5] Alexandra Chouldechova, A. Feder Cooper, Solon Barocas, Abhinav Palia, Dan Vann, and Hanna Wallach. Comparison requires valid measurement: Rethinking attack success rate comparisons in AI red teaming. In *Advances in Neural Information Processing Systems 38 (NeurIPS), Position Paper Track*, 2025. arXiv:2601.18076.
- [6] Julius Broomfield, Tom Ingebrechtsen, Esben Kran, et al. The structural safety generalization problem, 2025.
- [7] Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations, 2024.
- [8] James H. Kim and Anthony R. Scialli. Thalidomide: The tragedy of birth defects and the effective treatment of disease. *Toxicological Sciences*, 122(1):1–6, 2011.
- [9] Ross L. Prentice. Surrogate endpoints in clinical trials: Definition and operational criteria. *Statistics in Medicine*, 8(4):431–440, 1989.
- [10] Joshua Clymer, Nick Gabrieli, David Krueger, and Thomas Larsen. Safety cases: How to justify the safety of advanced AI systems, 2024.
- [11] Tomek Korbak, Joshua Clymer, Benjamin Hilton, Buck Shlegeris, and Geoffrey Irving. A sketch of an AI control safety case, 2025.
- [12] William R. Shadish, Thomas D. Cook, and Donald T. Campbell. *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Houghton Mifflin, Boston, 2002.
- [13] David Moher, Corinne S. Dulberg, and George A. Wells. Statistical power, sample size, and their reporting in randomized controlled trials. *JAMA*, 272(2):122–124, 1994.
- [14] James A. Hanley and Abby Lippman-Hand. If nothing goes wrong, is everything all right? interpreting zero numerators. *JAMA*, 249(13), 1983.
- [15] Teun van der Weij, Felix Hofstätter, Oliver Jaffe, Samuel F. Brown, and Francis R. Ward. AI sandbagging: Language models can strategically underperform on evaluations, 2025.

- [16] Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. XSTest: A test suite for identifying exaggerated safety behaviours in large language models, 2024.
- [17] Zhexin Zhang, Leqi Lei, Lindong Wu, Rui Sun, Yongkang Huang, Chong Long, Xiao Liu, Xuanyu Lei, Jie Tang, and Minlie Huang. SafetyBench: Evaluating the safety of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 15537–15553, 2024. arXiv:2309.07045.
- [18] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models, 2023.
- [19] Mantas Mazeika, Long Phan, Xuwang Yin, et al. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal, 2024.
- [20] Alexandra Souly, Qingyuan Lu, Dillon Bowen, et al. A StrongREJECT for empty jailbreaks, 2024.
- [21] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.
- [22] Leslie Kish. *Survey Sampling*. John Wiley and Sons, New York, 1965.
- [23] D. A. Williams. The analysis of binary responses from toxicological experiments involving reproduction and teratogenicity. *Biometrics*, 31(4), 1975.
- [24] Anthropic. Model card and evaluations for claude models. Technical report, Anthropic, July 2023.
- [25] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3982–3992, 2019.
- [26] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, et al. LMSYS-Chat-1M: A large-scale real-world LLM conversation dataset, 2024.
- [27] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012.
- [28] Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction, 2018.
- [29] Nicholas Carlini, Milad Nasr, Christopher A. Choquette-Choo, et al. Are aligned neural networks adversarially aligned?, 2024.
- [30] Xiaogeng Liu, Nan Xu, Muhao Chen, and Chaowei Xiao. AutoDAN: Generating stealthy jailbreak prompts on aligned large language models, 2024.

- [31] Maksym Andriushchenko and Nicolas Flammarion. Does refusal training in LLMs generalize to the past tense?, 2025.
- [32] Zheng-Xin Yong, Cristina Menghini, and Stephen H. Bach. Low-resource languages jailbreak GPT-4, 2024.
- [33] Anay Mehrotra, Manolis Zampetakis, Paul Kassianik, et al. Tree of attacks: Jail-breaking black-box LLMs automatically, 2024.
- [34] Fábio Perez and Ian Ribeiro. Ignore previous prompt: Attack techniques for language models, 2022.
- [35] OpenAI. GPT-4 technical report, 2023.
- [36] OpenAI. GPT-4o system card, 2024.
- [37] Anthropic. The claude 3 model family: Opus, sonnet, haiku. Technical report, Anthropic, March 2024.
- [38] Anthropic. System card: Claude opus 4 and claude sonnet 4. Technical report, Anthropic, May 2025.
- [39] Gemini Team, Google. Gemini: A family of highly capable multimodal models, 2023.
- [40] Gemini Team, Google. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context, 2024.
- [41] Hugo Touvron, Louis Martin, Kevin Stone, et al. Llama 2: Open foundation and fine-tuned chat models, 2023.
- [42] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The llama 3 herd of models, 2024.
- [43] Anthropic. Responsible scaling policy. Technical report, Anthropic, 2024.
- [44] Deep Ganguli, Liane Lovitt, Jackson Kernion, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned, 2022.
- [45] Ethan Perez, Saffron Huang, Francis Song, et al. Red teaming language models with language models, 2022.